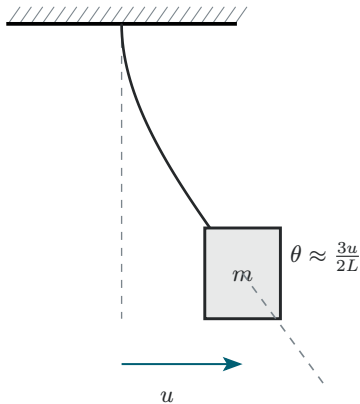
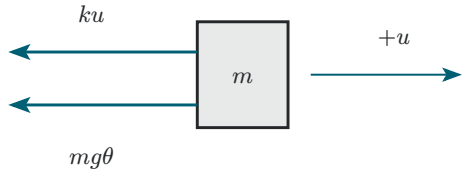


(a) Bending shape and tip slope



(b) Reduced restoring projection



Course-note approximation, PDF page 252.

$$m\ddot{u} + \left(k + \frac{3mg}{2L}\right)u = 0$$